

Child Disability and Parental Labor Supply

João Garcia, Rafael Latham-Proença, Marcela Mello and Joana Silva

May 2026

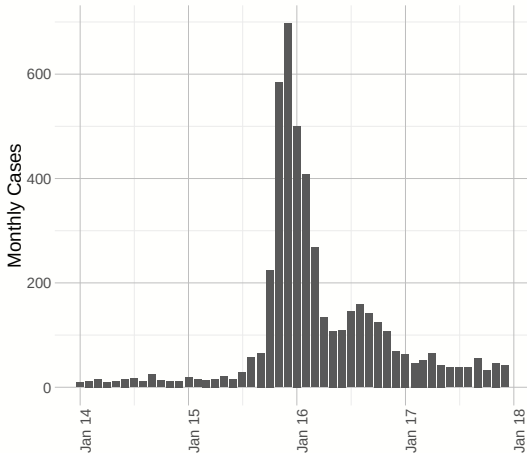
World Bank — LCR Economics Seminar Series

- Severe child disability creates a **double burden**
 - Parents need more time *and* more financial resources
- The burden falls disproportionately on **mothers**, amplifying the child penalty
- **Informal work**, prevalent among women in LAC, is widely assumed to offer flexibility for caregiving, but does it actually buffer against severe disability?
- Challenges: scarce data on informality; endogeneity of disability

- **Setting:** 2015 Zika outbreak in Brazil — a sudden, unexpected, exogenous shock to congenital disability
- **Design:** matched event-study around childbirth, using administrative data on **both formal and informal** labor
- **Findings, in brief:**
 - Mothers' formal employment falls **75%** more than the standard motherhood penalty
 - Equally large reductions in informal labor: *informality is not a buffer*
 - Disability transfers (BPC) raise total income but *amplify* employment reductions
 - No effects on family structure or subsequent fertility

Background

The Outbreak



- From near zero to peak in 3 months, in late 2015.
- Severe, lifelong disability requiring intensive caregiving
- Emergency declared in Nov 2015. By then, most infections had already happened.

Why It Works as Natural Experiment

- **No preventive behavior:**
 - CZS was previously unknown. Most infections happened *before* the public was aware
- **No selective abortion:**
 - Hard to diagnose in utero; Zika often asymptomatic; abortion heavily restricted
- **No direct effect on the mother:**
 - ~80% of adult cases asymptomatic, no lasting symptoms
- **Selective fertility unlikely:**
 - Outbreak public late 2015 → any fertility response shows up in births from mid-2016, when cases had already fallen

Data and Empirical Strategy

- **RESP-Microcefalia**
 - Surveillance registry of all probable/confirmed CZS cases
- **RAIS**
 - Universe of formal employment links, monthly outcomes
- **Single Registry** (*Cadastro Único*)
 - Administrative database covering bottom ~50% of population
 - Links datasets via tax ID;
 - Rare administrative source on *informal work* and *family structure*

Matching Strategy

- Need to control for **time**, **place** (virus exposure), and **individual risk factors**
- Exact matching on:
 - a) Municipality of residence (microregion in stage 2)
 - b) Month of birth
 - c) Maternal age
 - d) Education (high school completion)
 - e) Parity (firstborn)
 - f) Race (white)
- Variables informed by epidemiological literature and balance checks
- Final sample: **867 cases, 7,489 controls**

Matching details

Balance

Unmatched cases

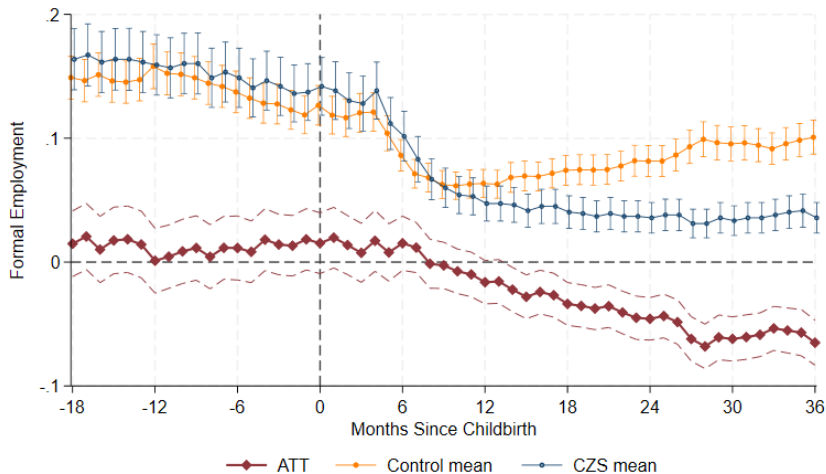
Fully saturated event-time model:

$$y_{i\tau} = \sum_{k \in \kappa} \mu_k 1\{\tau = k\} + \sum_{k \in \kappa} \beta_k 1\{\tau = k\} \cdot T_i + \varepsilon_{i\tau}$$

- μ_k : average outcome at k for control; β_k : difference for treated
- Weights: treated = 1; control = $n_t(s)/n_c(s)$ (stratum-balancing)
- Estimands: time series, simple post-period difference, DiD (with pre-period reference)
- Errors clustered at the stratum level

Main Results

Mothers' Formal Employment



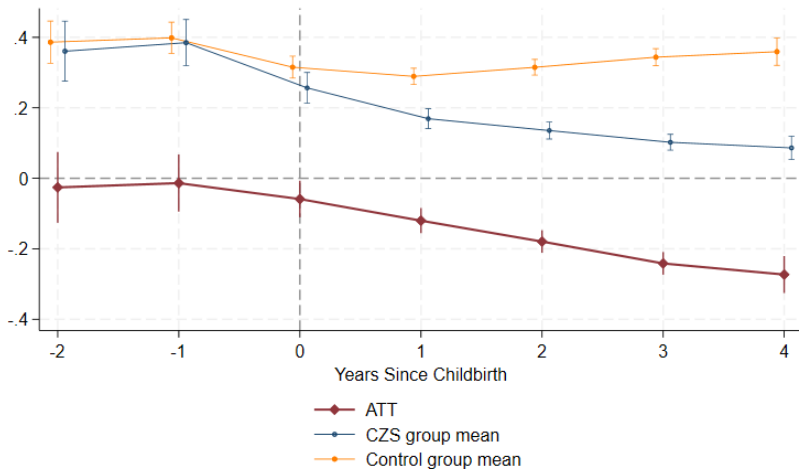
Mothers' Formal Labor Market

Variable	SD	DID	Baseline	N	Clusters
Employment	-0.040*** (0.007)	-0.051*** (0.013)	0.149	509,716	852
Earnings	-39.47*** (8.92)	-52.05*** (14.89)	155.4	509,716	852
Monthly wages if employed	83.29 (72.17)	79.47 (77.54)	1043.7	54,233	596
Weekly hours	-2.02*** (0.32)	-2.41*** (0.56)	6.68	509,716	852
Weekly hours if employed	-0.68 (0.83)	0.72 (1.09)	42.6	59,659	599

Disability penalty: **12.0 p.p.**, **75% larger** than the standard motherhood penalty (6.8 p.p. for controls).

Earnings / hours figures

Mothers' Informal Employment



Mothers' Informal Labor Market

Variable	SD	DID	Baseline	N	Clusters
Employment	-0.204*** (0.015)	-0.190*** (0.041)	0.399	29,979	852
Earnings	-384.6*** (45.9)	-493.57*** (132.719)	690.9	29,979	852
Conditional months of work	0.010 (0.473)	0.641 (0.663)	7.81	9,984	647
Conditional earnings	55.5 (259.1)	-870.4 (779.4)	1779.3	9,984	647

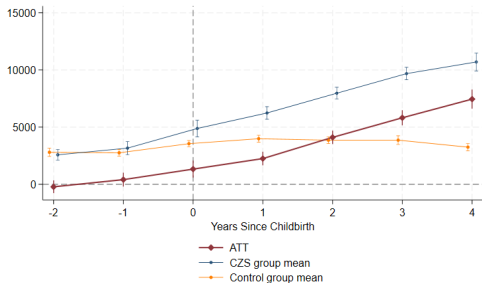
Effect grows over time: **12 p.p.** in year 1, **27 p.p. by year 4**. Informal labor offers no buffer against intensive caregiving demands.

Earnings / months figures

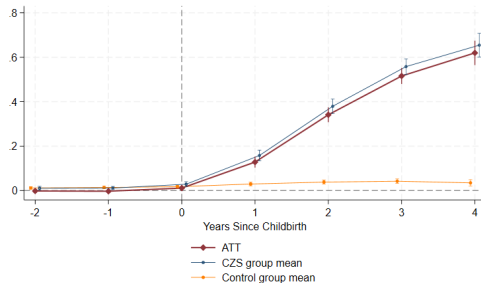
Income and Social Assistance

Total Household Income and BPC Take-up

Total Annual Family Income



Probability of Receiving BPC



Despite labor income falling, **total income rises** — BPC take-up climbs from ~ 0 to **60%** by year 4.

Income Sources and Expenses



Pension transfers more than offset lost labor income; expenses rise on food (formula feeding) and medicine. Transfers cover *real* additional costs.

Does the BPC Drive the Employment Reduction?

By year 4, **60%** of CZS families receive BPC (1 min. wage, means-tested)

Causal

BPC induces additional reduction

- income effect
- income-test kink

Marker

BPC flags families with larger reductions

- severity
- low baseline employment

Identification — two layers of selection on observables:

- *CZS effect*: matching strategy
- *BPC heterogeneity*: flexibly control for Propensity Score of BPC receipt

Falsification: pre-birth employment → null.

PS estimation

Effects on Formal Employment by BPC Recipient Status

	(1)	(2)	(3)	(4)
Outcome: Formal Employment 2 Years After Childbirth				
CZS	-0.077*** (0.016)	-0.069*** (0.016)	-0.066*** (0.019)	-0.068*** (0.016)
BPC	-0.017 (0.017)	0.006 (0.020)	0.006 (0.020)	0.010 (0.020)
CZS x BPC	-0.037* (0.022)	-0.060** (0.024)	-0.060** (0.024)	-0.063** (0.025)
PS Controls	None	Linear	Cubic	Ventile FE
PS Controls x CZS	-	Yes	Yes	Yes
Observations	8,356	8,354	8,354	8,354
Clusters	852	851	851	851

Non-recipients: -6.9 p.p.; recipients: **-12.9 p.p. (nearly double)**. Naive spec biased *toward zero*.

Placebo

Effects on Informal Employment by BPC Recipient Status

	(1)	(2)	(3)	(4)
Outcome: Informal Employment 2 Years After Childbirth				
CZS	-0.098*** (0.024)	-0.100*** (0.024)	-0.075** (0.030)	-0.096*** (0.024)
BPC	0.016 (0.023)	-0.001 (0.025)	0.001 (0.025)	0.002 (0.026)
CZS x BPC	-0.190*** (0.031)	-0.183*** (0.034)	-0.188*** (0.035)	-0.199*** (0.035)
PS Controls	None	Linear	Cubic	Ventile FE
PS Controls x CZS	-	Yes	Yes	Yes
Observations	7,494	7,492	7,492	7,492
Clusters	845	844	844	844

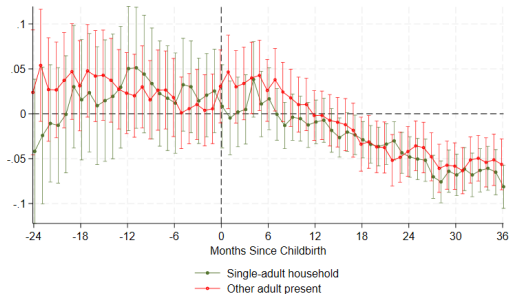
Recipients: **-28.8 p.p.**; non-recipients: -9.8 p.p. — **~3× larger** interaction than in formal work.

Placebo

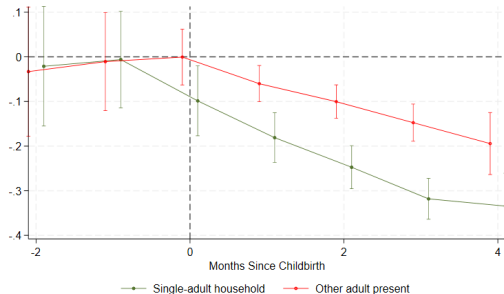
Heterogeneity

Heterogeneity by Family Structure

Effect on Mother's Formal Employment



Effect on Mother's Informal Employment



Single mothers show larger and earlier reductions in informal employment, but not in formal employment.

Family structure (null)

Severity

Wrap-up

External Validity: Where Should These Findings Apply?

Dimension	Study context	Generalizes to	Caveat
Severity	Severe lifelong CZS	Severe CP, profound autism	Smaller effects for moderate disabilities
Population	Low-income Cadastro Único families	Median households in LMICs (India, Indonesia)	Wealthier households have more care substitutes
Labor market structure	40% informal / 15% formal	High-informality settings (most of LAC); 61% of global employment informal	Informal flexibility may help less intensive care needs
Transfers	BPC: high takeup, relatively generous	Most LMICs: only 19% of disabled receive payments	Without transfers: smaller effects but greater financial pressure on women

Other Analyses in the Paper

- **Robustness**

- Reweighting for unmatched cases → main estimates nearly unchanged ▶
- Mortality Lee bounds → exclude zero for employment, earnings, hours ▶
- SR-data missingness → CZS not a significant predictor of SR presence ▶
- BPC placebos on pre-birth employment → null ▶

- **Other outcomes**

- Family structure & fertility → no effects on separation, or subsequent fertility ▶
- Wages, hours conditional on employment → no intensive-margin adjustment ▶
- Fathers' labor supply → null on formal, negative on informal (small sample) ▶

- Mothers of children with severe disability see strong, persistent labor market exit
- **Informality is not a buffer**, challenging conventional wisdom
- Disability transfers raise income substantially but **amplify** employment reductions

Policy Implications

- Disability-transfer programs face a **tradeoff**: income protection vs. work disincentives, especially for mothers
- Alternative designs worth exploring:
 - Benefits varying with employment status
 - In-kind care services that *enable* maternal employment
- Open question for LAC: minimum bundle of services that improves child outcomes *and* supports caregivers' labor force participation
- Existing infrastructure (community health workers) as delivery channels

Appendix

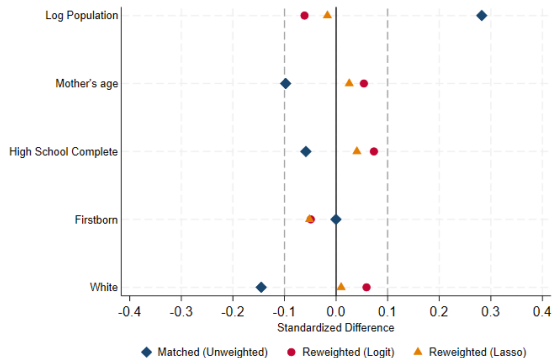
Sample Construction and Matching Flow

Stage	Cases
<i>A. Initial Sample Construction</i>	
RESP records (probable/confirmed CZS)	3,594
After dropping missing birth dates	2,946
After restricting to 2015 – 2017	2,431
After de-duplication	2,411
SR-2017 linkage	940
SR-2019 linkage	281
Dropped due to missing data	21
Cases available for matching	1,200
<i>B. Sequential Matching Process</i>	
Stage 1: Same Municipality	669 (6,009 controls)
Stage 2: Same Microregion	198 (1,480 controls)
Final sample (cases)	867
Final sample (controls)	7,489

Balance Table

	Microcephaly	Control (raw)	Control (wgt)	p-value
Completed HS (%)	37.37	29.06	37.37	.
Mother's age	24.64	23.08	24.64	.
White	16.84	11.20	16.84	.
Black	11.53	16.33	12.96	0.254
Yellow	0.58	0.51	0.66	0.788
Pardo	70.47	71.76	69.19	0.316
Indigenous	0.58	0.21	0.35	0.344
Firstborn	57.44	60.46	57.44	.
Second	27.22	24.05	26.87	0.767
Third	9.57	10.60	10.18	0.580
Higher	5.77	4.89	5.51	0.765
Employment y-1	14.92	13.17	13.87	0.371
Employment y-2	16.29	14.22	14.70	0.171
Wage y-1	1083.63	990.32	1044.88	0.552
Wage y-2	1062.34	952.13	987.27	0.163

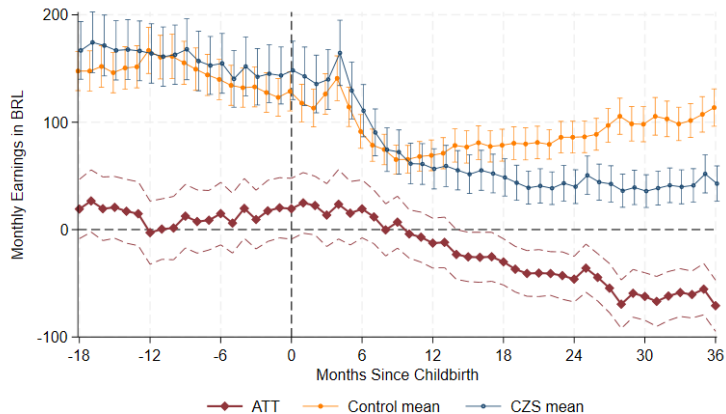
Robustness: Unmatched Cases



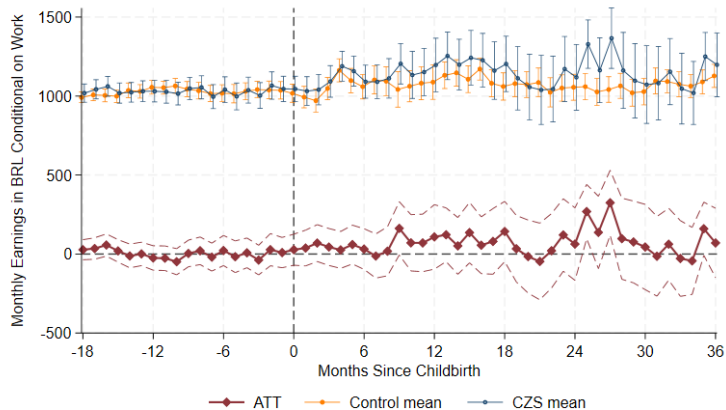
- 867 of 1,200 SR cases find matched controls
- Reweight to recover the full SR sample (Stuart 2011)
- Matched sample shows selection on race, age, microregion size; reweighting eliminates these gaps
- Reweighted main estimates nearly identical to baseline

Formal Employment Figures

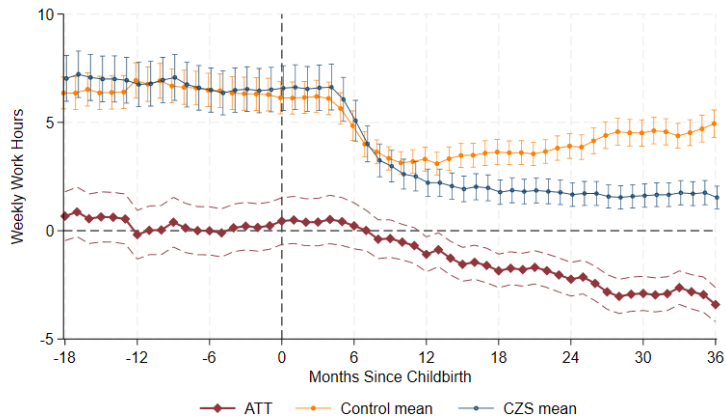
Mothers' formal earnings



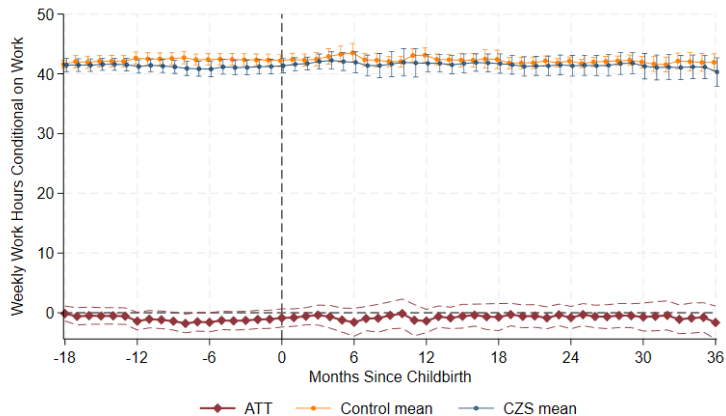
Mothers' wages if employed



Weekly contractual hours

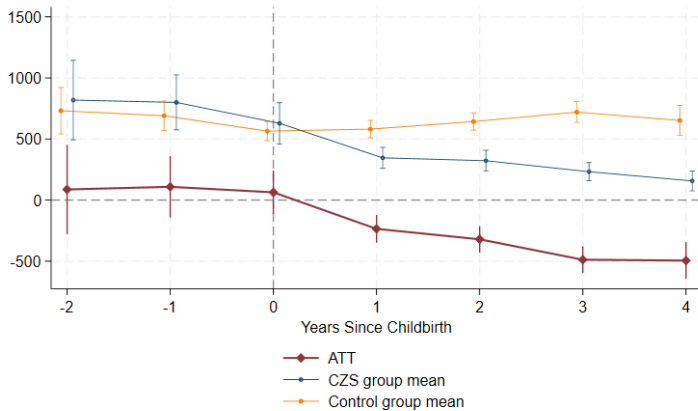


Weekly contractual hours if employed

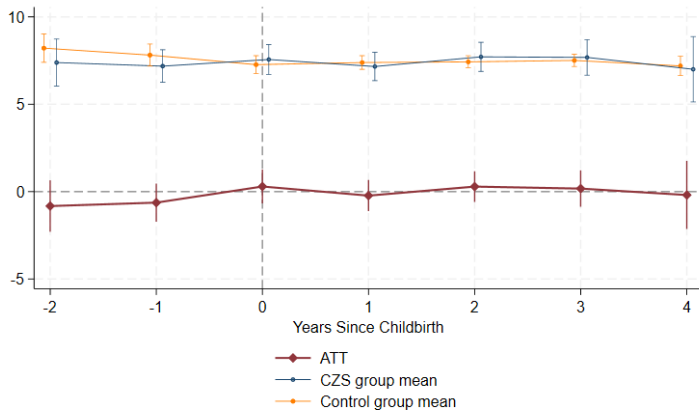


Informal Employment Figures

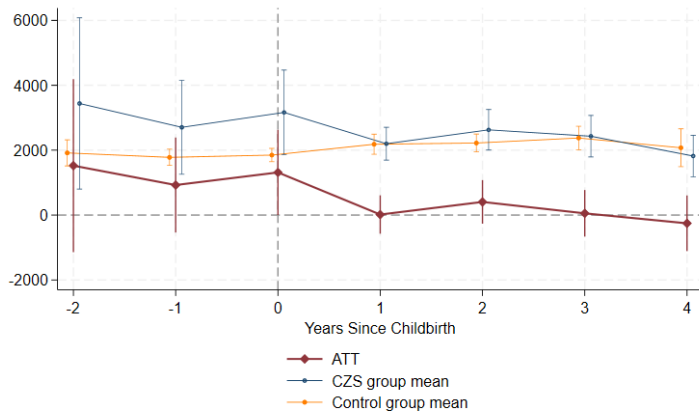
Mothers' informal earnings



Conditional months of work



Conditional earnings



Placebo: Effects on Formal Employment by BPC Status

	(1)	(2)	(3)	(4)
Placebo: Formal Employment 1 Year Before Childbirth				
CZS	-0.017 (0.020)	-0.016 (0.020)	0.012 (0.024)	-0.020 (0.020)
BPC	-0.061*** (0.021)	-0.031 (0.023)	-0.030 (0.023)	-0.024 (0.022)
CZS x BPC	0.035 (0.031)	0.030 (0.033)	0.026 (0.033)	0.027 (0.032)
PS Controls	None	Linear	Cubic	Ventile FE
PS Controls x CZS	-	Yes	Yes	Yes
Observations	8,356	8,354	8,354	8,354
Clusters	852	851	851	851

Placebo: Effects on Informal Employment by BPC Status

	(1)	(2)	(3)	(4)
Placebo: Informal Employment 1 Year Before Childbirth				
CZS	0.009 (0.056)	0.011 (0.056)	0.037 (0.072)	-0.016 (0.055)
BPC	-0.016 (0.046)	-0.042 (0.048)	-0.039 (0.048)	-0.047 (0.049)
CZS x BPC	-0.054 (0.083)	-0.057 (0.089)	-0.060 (0.089)	-0.044 (0.091)
PS Controls	None	Linear	Cubic	Ventile FE
PS Controls x CZS	-	Yes	Yes	Yes
Observations	1,921	1,921	1,921	1,921
Clusters	365	365	365	365

BPC Propensity Score: Average Marginal Effects

Severe Microcephaly	0.106*** (0.032)
Race: White	-0.014 (0.046)
Mother completed high school	-0.037 (0.038)
Mother's year of birth	-0.001 (0.003)
Mother had a formal job in y-1	0.001 (0.041)
Income per person in y-1	-0.000 (0.000)
Not in Single Registry in y-1	0.046 (0.044)
Birth Order:	
Second	0.074 (0.043)
Third	0.064 (0.063)
Fourth	0.044 (0.085)
Fifth or higher	0.229 (0.155)
Observations	866

Back

Family Figures

Family Structure: Null Effects

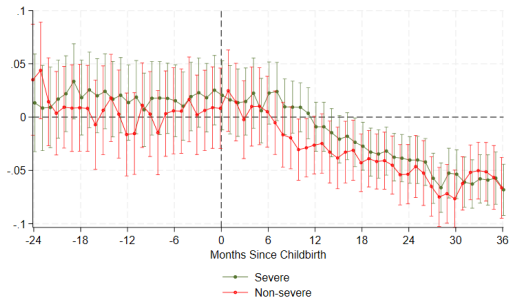
Variable	SD	DID	Baseline	N	Clusters
Single household	0.005 (0.020)	0.014 (0.034)	0.504	26,782	844
Spouse present	0.051*** (0.018)	-0.004 (0.031)	0.253	26,782	844
Grandparent present	-0.044*** (0.011)	-0.014 (0.018)	0.178	26,782	844
Subsequent fertility	0.009 (0.010)	0.009 (0.010)	0	26,782	844

No effect on cohabitation with spouse, grandparents, or subsequent fertility. The vulnerable population in our sample shows resilient family structure.

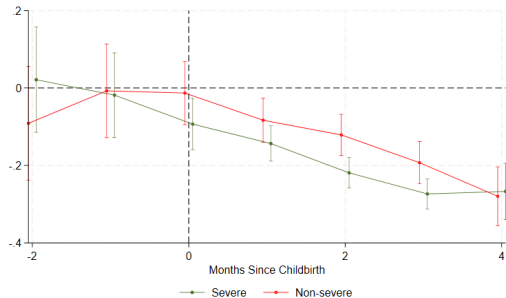
[Figures](#)[Back](#)

Heterogeneity by Severity

Effect on Mother's Formal Employment

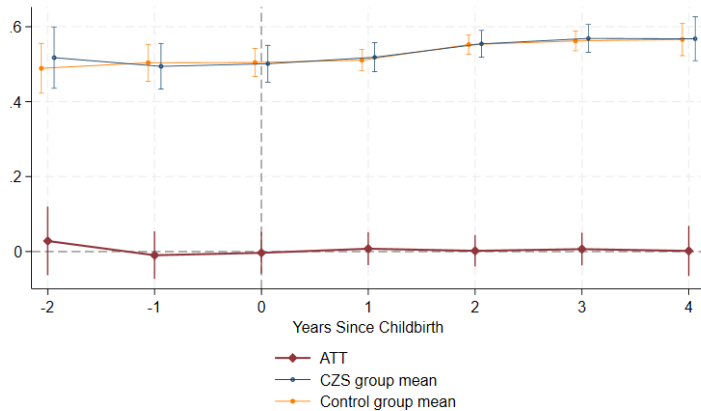


Effect on Mother's Informal Employment

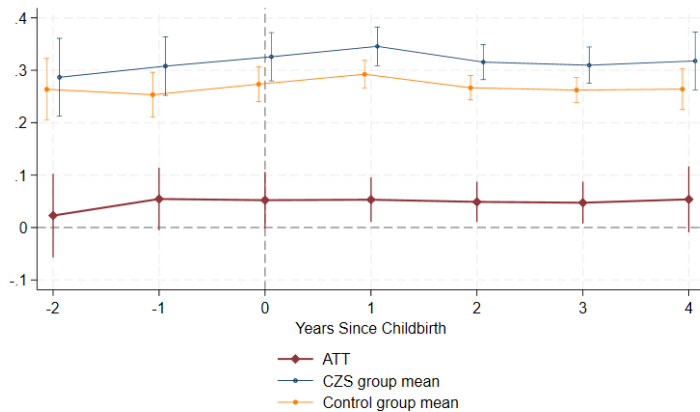


Severe microcephaly (head circumference >3 SD below mean) drives faster exit from informal work; convergence by year 4. Even non-severe CZS requires intensive caregiving.

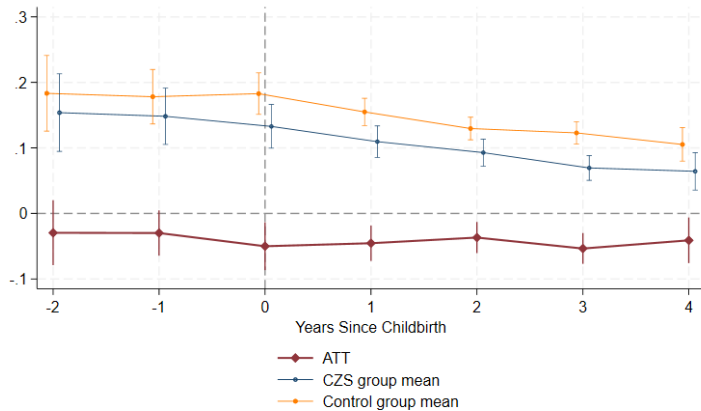
Mother lives alone



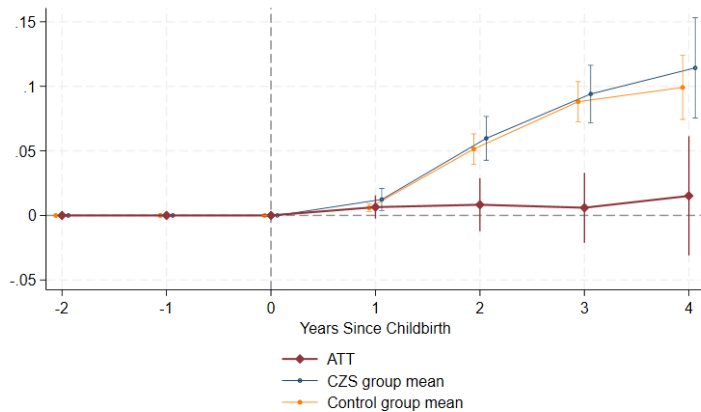
Father cohabits



Grandparents present

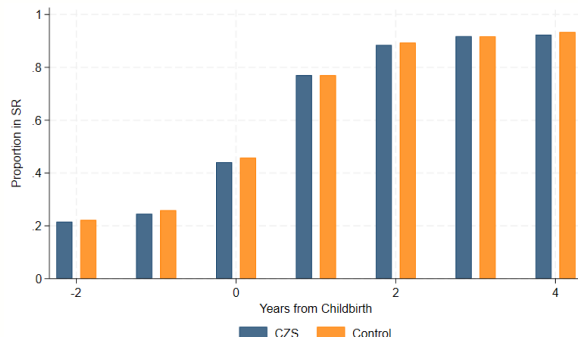


Subsequent fertility



SR Missingness

SR Data Missingness



- Few families in SR before childbirth (~20–25%); rises to ~90% by year 2 (eligibility kicks in around childbirth)
- Patterns virtually identical for CZS cases and controls

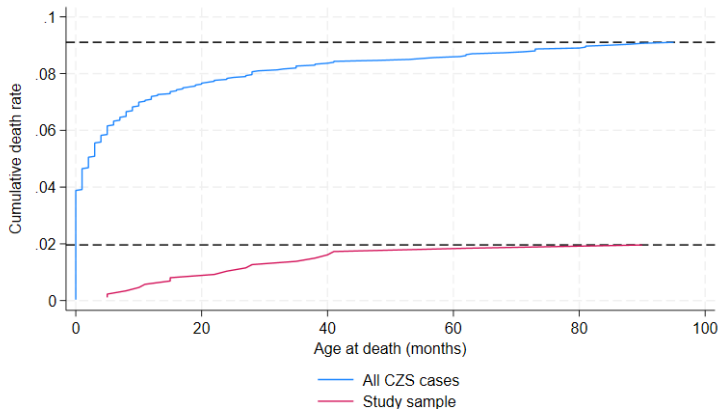
AME of CZS on probability of being in SR

Time from Childbirth	Unadjusted		Demographi
	Diff. (p.p)	p-value (%)	Diff. (p.p)
-1 year	-1.3	0.53	-1.4
Birth	-1.7	0.47	-1.8
+1 year	0.0	0.99	-0.0
+2 years	-0.9	0.57	-0.8
+3 years	0.1	0.97	0.1

All AMEs < 2 p.p. and not statistically significant: CZS does not predict SR presence.

Mortality

Mortality in Sample vs Full RESP



Sample mortality is much lower than the full RESP — selection toward surviving children. Lee bounds on next slide.

Lee Bounds for Mortality Sensitivity

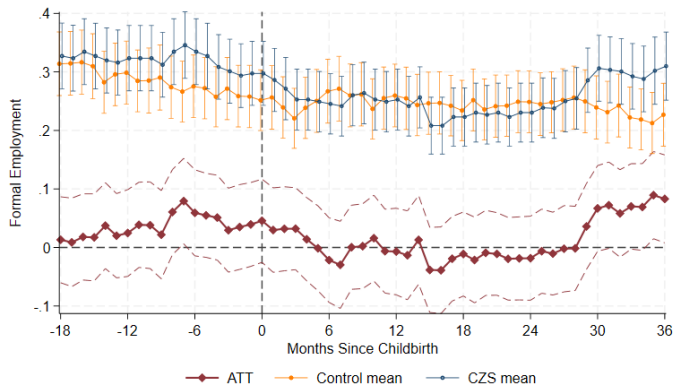
Formal Sector	Min	Max
Employment	-0.041	-0.032
Earnings	-40.190	-21.812
Monthly wages if employed	-130.401	251.871
Weekly hours	-2.069	-0.819
Weekly hours if employed	-7.993	15.774

Informal Sector	Min	Max
Employment	-0.202	-0.190
Earnings	-436.847	-320.940
Conditional months of work	0.017	0.531
Conditional annual earnings	59.043	1187.402

Bounds exclude zero for employment, unconditional earnings, and hours.

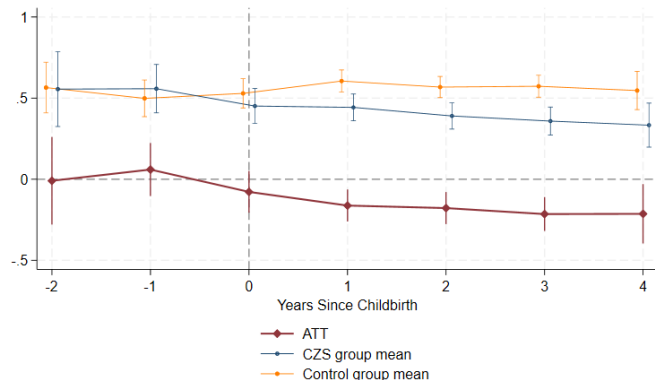
Father Results

Fathers: Formal Employment



Null effect on formal employment. Sample limited to cohabiting fathers (~25% of households).

Fathers: Informal Employment



Significant reduction in informal employment, suggesting compositional exit of marginally-attached fathers.